

Section A

Answer **all** questions in the spaces provided.

Refer to the separate Resource Folder to answer question 1.

1. (a) (i) The Red Admiral butterfly competes with greenfly for food. The scientific name for one type of greenfly is *Acyrtosiphon pisum*.

Complete the table below to show the scientific classification of greenfly. [3]

Kingdom:	
Phylum:	
Class:	Insecta
Order:	Hemiptera
Family:	Aphididae
Genus:	
Species:	
Scientific name:	<i>Acyrtosiphon pisum</i>

- (ii) Another butterfly is found in Africa. Its scientific name is *Vanessa abyssinica*.



- I. State **one** advantage of using scientific names. [1]

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- II. State **one** piece of evidence, from its scientific name, that shows it is related to the Red Admiral. [1]

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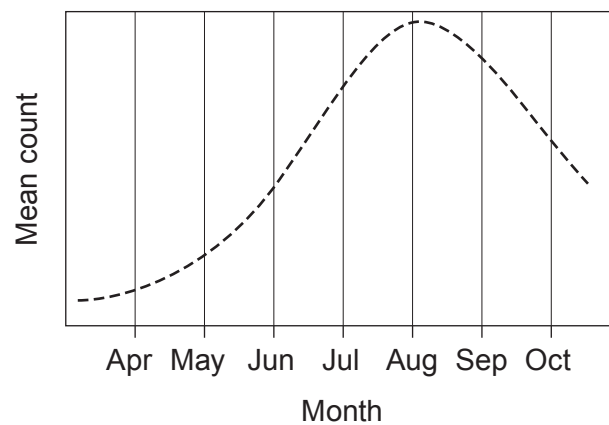
(b) The variation in population of adult Red Admirals is shown by the graph in **Figure 1**.

- (i) I. State the name of the seasonal movement of animals as seen in Red Admirals. [1]

II. State **one** reason why this seasonal movement occurs. [1]

- (ii) Use the information in the graph in **Figure 1** to compare the variation of the Red Admiral population in 2019 with the mean change since 1976. [2]

- (iii) Red Admirals are a food source for frogs. **Add a line** to the graph below to show how the population of frogs changed between April and October in 2019. [2]



- (iv) Use your line to explain the relationship between numbers of predators (frogs) and the numbers of their prey (Red Admirals). [3]



- (c) (i) Use the information in **Table 2** to construct the part of the food web that depends on berries. [3]

Examiner
only

Berries



(ii) Explain why the frog can be found at more than one trophic level. [2]

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(d) Refer to **Table 2** to construct a pyramid of biomass for a food chain of **five** trophic levels starting with berries and ending with buzzards. [3]

(e) Use the information in **Figure 2** to calculate the efficiency of energy transfer between the primary and secondary consumers. [3]

% efficiency =

